

(TR-104) PYTHON WITH AI/ML

Training Day 59 Report

Date: 2 April 2026

Topic Covered: Social Media Data Extraction, Trend Analysis & Automation Tools

The fifty-ninth day of training focused on website testing validation, social media data extraction using Apify, and building a multi-source trend analysis pipeline. The session also included testing automation tools like Playwright and comparing scraping approaches for real-world reliability.

- Tested contact and enquiry forms and verified successful submission and proper response handling.
- Confirmed that previously identified form issues were resolved and system behavior is stable.
- Implemented Instagram hashtag data extraction using Apify Python SDK.
- Used apify/instagram-hashtag-scraper actor for extracting posts related to “trendingreels”.
- Configured scraping parameters including hashtags list and result limits for controlled data extraction.
- Extracted structured fields such as username, caption, hashtags, likes, comments, post type, and URL.
- Cleaned and processed hashtag data for consistency and removed invalid entries.
- Built a structured dataset for further trend analysis and insights generation.
- Developed a trend analysis pipeline integrating Google Trends, Reddit RSS, and YouTube data sources.
- Identified Reddit API limitations due to missing credentials and used RSS feed as an alternative.
- Successfully retrieved trending data from YouTube and Reddit RSS feeds.
- Generated trending keywords and identified top-performing posts across platforms.
- Tested Instagram automation using Playwright and observed login restrictions preventing direct scraping.

- Compared Playwright with Apify and concluded Apify provides more stable and structured data access.
- Strengthened understanding of real-world data collection challenges and multi-source trend analysis systems.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/trends>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/fetching-instagram-hashtag-trends-using-apify-python-08edf71bff80>